

INTERPRETASI DATA

Tujuan Interpretasi Data:

- *Untuk menjabarkan data hasil uji.*
- *Menelusuri penyebab ketidak normalan*
- *Mendukung ketepatan action plant*

Interpretasi data particle count.

Example Particle Count (about 1 ppm)	
Size (microns)	Count Larger Than Size per ml
1	1752
2	1396
5	517
10	144
15	55
20	25
50	1.3
75	0.27
100	0.08

TABLE 1

$R_2/R_5/R_{15}$
currently part
of ISO Standard

517 particles > 5µm/ml

55 particles > 15µm/ml

ISO 16/13

XX/YY/ZZ

XX : 4 - 5 µm

YY : 6 - 13 µm

ZZ : > 14 µm

Number of particles per mL		Range Number (R)
More than	Up to and including	
80,000	160,000	24
40,000	80,000	23
20,000	40,000	22
10,000	20,000	21
5,000	10,000	20
2,500	5,000	19
1,300	2,500	18
640	1,300	17
320	640	16
160	320	15
80	160	14
40	80	13
20	40	12
10	20	11
5	10	10
2.5	5	9
1.3	2.5	8
0.64	1.3	7
0.32	0.64	6
0.16	0.32	5
0.08	0.16	4
0.04	0.08	3
0.02	0.04	2
0.01	0.02	1

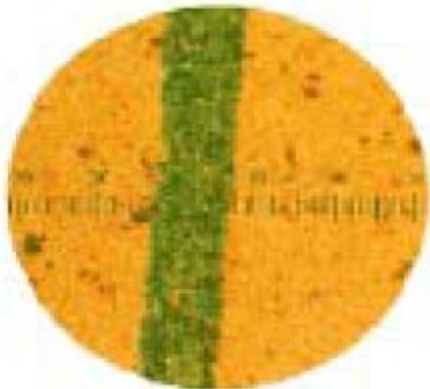
TABLE 2

Penulisan dalam laporan cukup : YY/ZZ, karena partikel dengan ukuran 4-5 µm tidak mempengaruhi kinerja filter

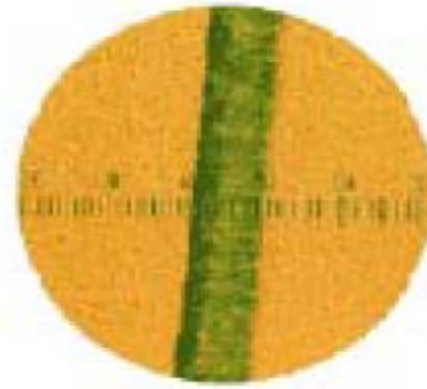
Kriteria cleanliness ISO 4406

	12/9		14/11		16/13		18/15		20/17		22/19		24/21		26/23
Hydraulic fluids	Very Clean			Clean					Dirty						
Gear Oils				Very clean			Clean								Dirty
Engine Lubes			Very clean			Clean					Dirty				
Turbin Oils		Very clean			Clean		Dirty								

Minyak pada kondisi 19/14
ISO 4406



Minyak pada kondisi 10/17
ISO 4406



Contoh :

Analisa minyak pelumas turbin.

Dalam laporan tertulis : Cleanliness ISO 4406 : 21/16

Artinya :

Minyak pelumas tersebut mengandung :

10.000-20.000 partikel /ml dengan ukuran 6-13 um dan

320-640 partikel / ml dengan ukuran 15-55 um

Standard ISO 4406 untuk minyak turbin : 17/15/12

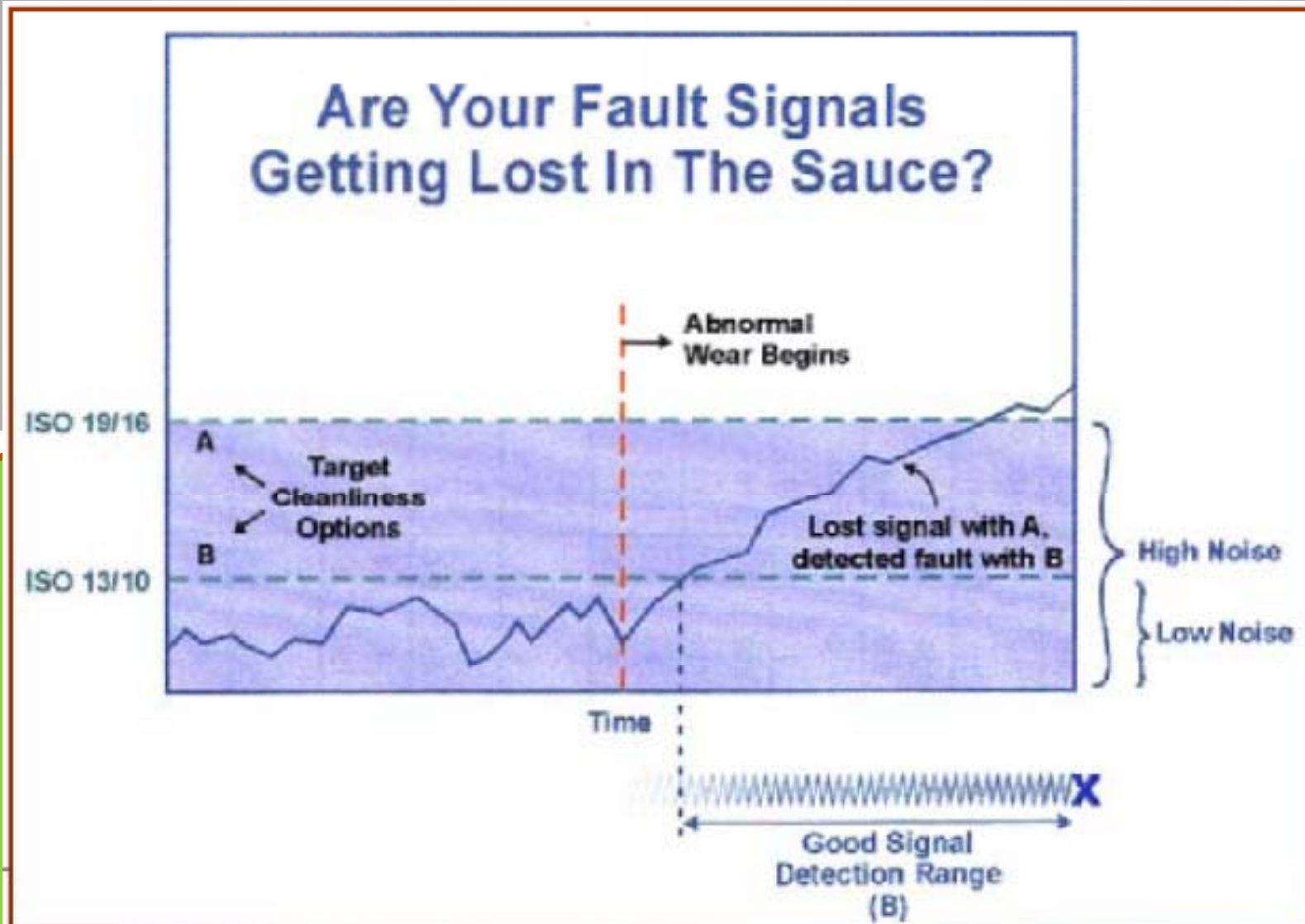
160-320 partikel /ml dengan ukuran 6-13 um dan

20-40 partikel / ml dengan ukuran 15-55 um

Kesimpulan : Cleanliness hasil analisa **Out of range**

Rekomendasi : Lakukan filtrasi, cek kondisi filter, lihat prosedur diagnosa.

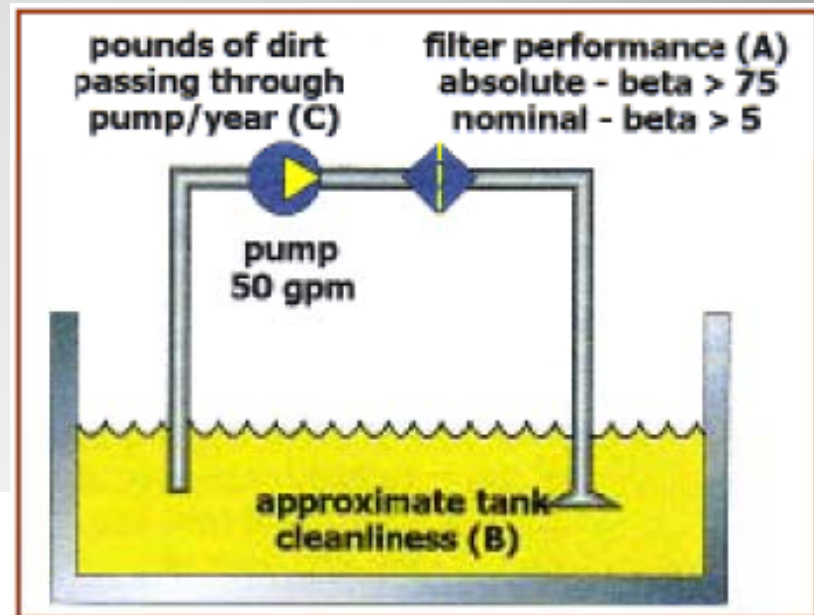
Efek Tidak Tepatnya Menentukan ISO Baseline



Contoh : Pada system hydraulic oil

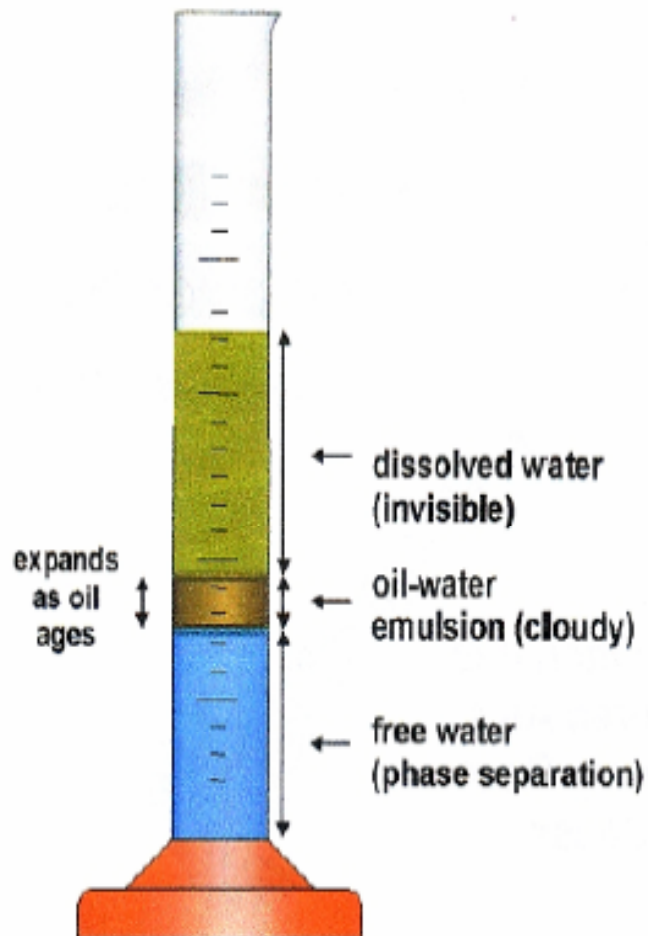
Filter (A)	ISO Code (B)	Dirt (lbs.) (C)	50 lb. bags	Relative pump life
25 micron nominal	21/18	6784	136	1
10 micron nominal	19/16	1809	36	1.9
10 micron absolute	16/13	211	4.2	4.4
6 micron absolute	14/11	53	1	8.8
3 micron absolute	12/9	14	0.28	15

All figures are approximations.



- Dengan menggunakan ISO 21/18, sistem hidraulic ini memompa 136 karung (ukuran ± 25 kg) kotoran setiap tahun
- Dengan menggunakan ISO 14/11, hanya 1 karung (ukuran ± 25 kg) kotoran setiap tahunnya yang melewati pompa

Water content.



- **Dissolved Water:** water molecules are dispersed one-by-one through-out the oil. Like humidity.
- **Emulsified Water:** microscopic globules of water are dispersed in stable suspension in the oil. Like fog.
- **Free Water:** water, when mixed with oil, readily settles to tank/sump bottom. Like rain.

Oil	Dissolved* (ppm)	Emulsified* (ppm)	Free* (ppm)
New Hydraulic Fluid	0-200	200-1000	>1000
Aged Hydraulic Fluid	0-500	500-5000	>5000
New R&O Oil	0-150	150-500	>500
Aged R&O Oil	0-500	500-1000	>1000
New Crankcase Oil	0-2000	2000-5000	>5000

*approximate values

Water Concentration, ppm	Status	Action
100 - 300	Alert	Check seals, breathers, coolers, etc., for ingress sources; watch trend
300 - 800	Danger	Aggressively investigate and correct source of ingress and implement effective water removal activities
800+	Extreme Danger	Immediate action is required to eliminate ingress and effect removal of water to minimize damage to machine and lubricant

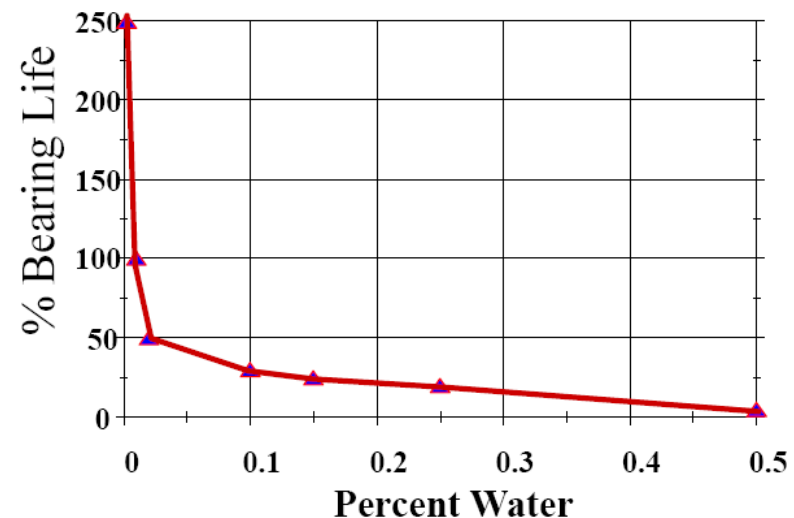


WATER

Water in oil can increase oxidation rate by more than ten times

Water and Bearing Life

Based on 100% Life at 0.01% Water



Reference: Timken Bearing Company

Water Related Problem

Hydrolysis and Oxidation



Water promotes chemical and physical changes in the base oil.

- Acid formation
- Oil thickening
- Varnish
- Sludge

Aeration



Water encourages aeration problems.

- Foaming
- Air entrainment

Viscosity Effects



Water leads to stable emulsions, higher viscosity and non-newtonian properties.

Dielectric Effects



Water reduces electric insulating properties of oil.

Water Related Problem

Corrosion



Water in oil gives acid greatest destructive potential.
Water is the central promotor of rust.

Film Strength Loss



Water in elastohydrodynamic contacts (e.g., rolling element bearings) cause film strength failure and hydrogen embrittlement.

Water flash-vaporization in hydrodynamic contacts causing bearing failure.

Cavitation



Water is the leading cause of hydraulic pump cavitation (vaporous cavitation).

Silting and Filter Plugging



Water causes poor filterability and valve stiction.

Elemental Spectroscopy (AAS)

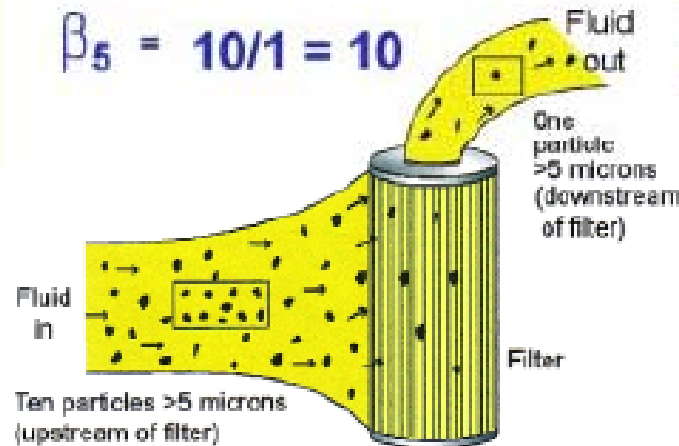
Katagori sumber element (Wear, Contaminant, Additives)

Element	Wear	Contamination	Additive
Iron (Fe)	X	X	
Copper (Cu)	X	X	X
Chromium (Cr)	X		
Tin (Sn)	X		
Aluminum (Al)	X	X	
Lead (Pb)	X		
Silicon (Si)		X	X
Sodium (Na)		X	X
Boron (B)		X	X
Calcium (Ca)		X	X
Magnesium (Mg)		X	X
Zinc (Zn)	X		X
Phosphorous (P)		X	X
Molybdenum (Mo)			X
Potassium (K)		X	

Beta Rating

$$\beta_x = \frac{\text{No. of particles greater than X microns upstream}}{\text{No. of particles greater than X microns downstream}}$$

$$\beta_5 = 10/1 = 10$$



Multipass Test Stand

% Efficiency =

$$\frac{\beta - 1}{\beta} \times 100$$

Contoh

$\beta_4 = 1,000,000 / 5,000 = 200$; Filter efficiency would be $= (1 - 1 / 200) \times 100\% = 99.5\%$

In this case, one million upstream particles 4 microns and larger passing through a beta 200 filter would have a downstream count of 5,000 particles 4 microns and larger for a 99.5% filtration efficiency.

Contoh

$$\beta_4 = 200 / 100 = 2$$

4 = the particle size of interest is 4 microns

200 = the number of particles 4 microns and larger that represent the upstream or current particle count

100 = the number of particles 4 microns and larger that represent the downstream or desired particle count

2 = the resulting beta ratio, β

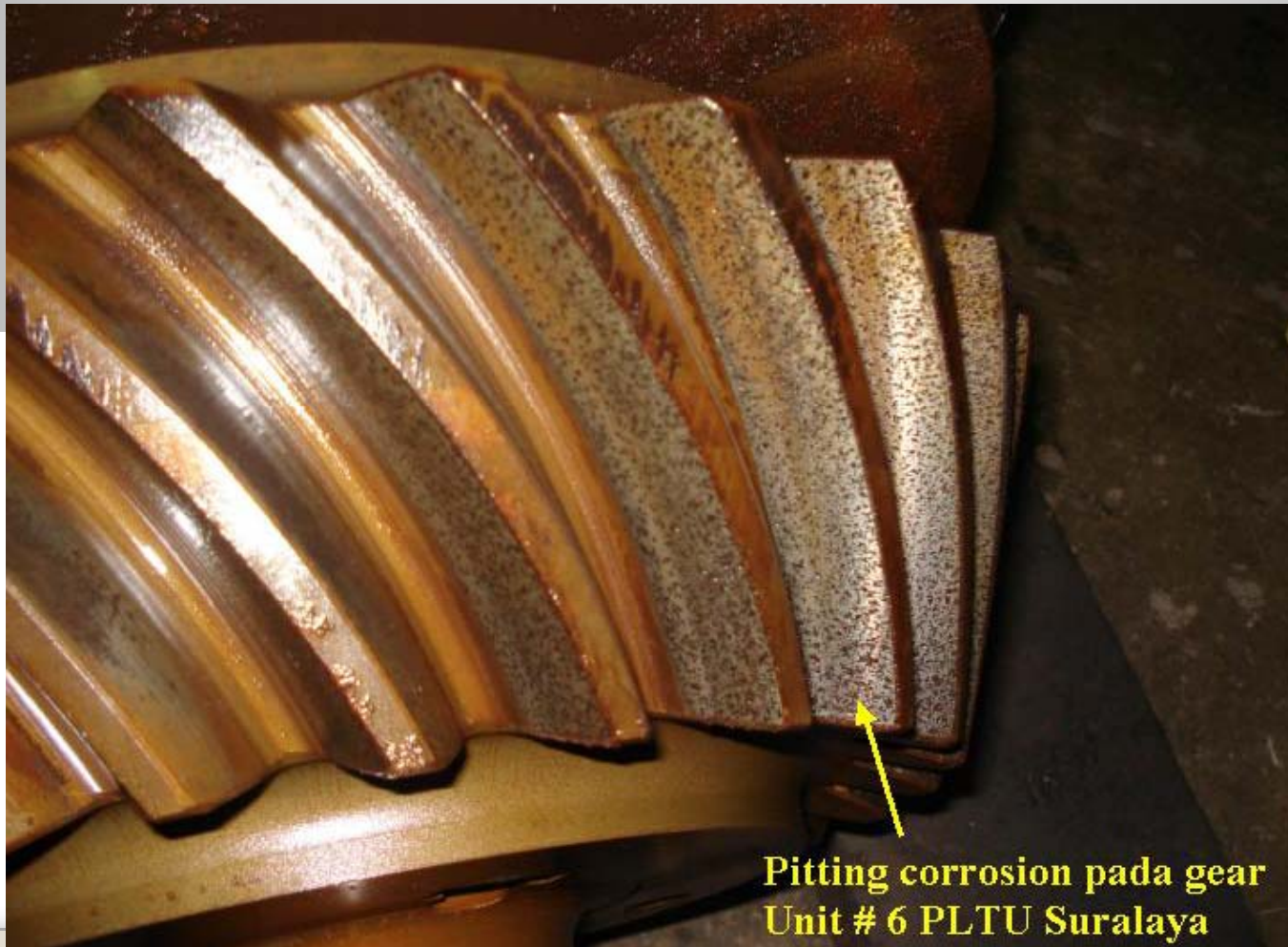
$$\text{Filter efficiency} = (1 - 1 / \beta_x) \times 100\%$$

if $\beta = 2$, filter efficiency = 50%.

STUDY KASUS

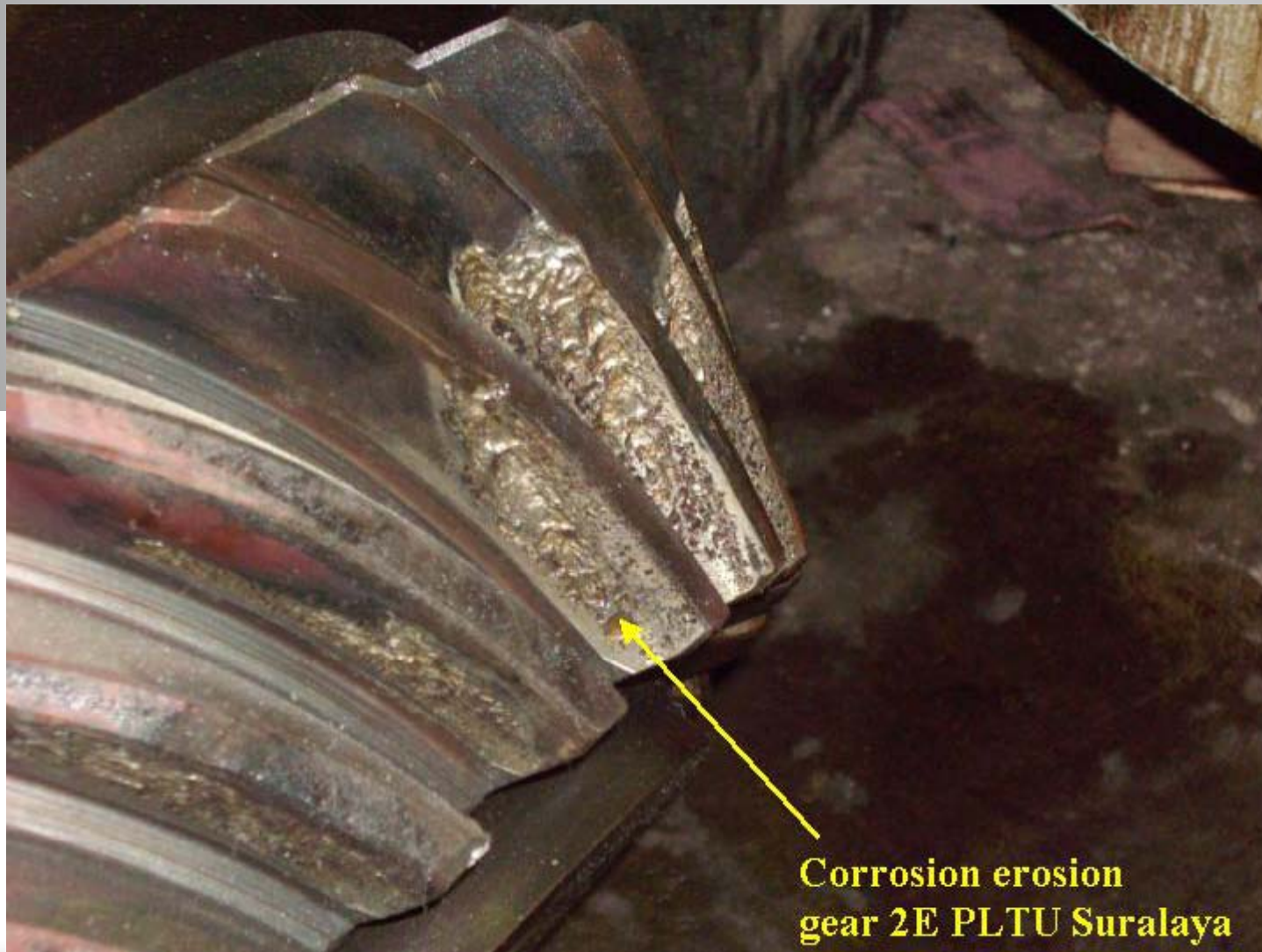
CONTOH KASUS KERUSAKAN AKIBAT MINYAK PELUMAS (KASUS SEBELUM DILAKUKAN PDM)

Pinion gear rusak karena pitting



Pitting corrosion pada gear
Unit # 6 PLTU Suralaya

CONTOH KASUS KERUSAKAN AKIBAT MINYAK PELUMAS



**Corrosion erosion
gear 2E PLTU Suralaya**

CONTOH KASUS KERUSAKAN AKIBAT MINYAK PELUMAS

Pitting corrosion pada gear
Unit #1C PLTU Suralaya



Gambar Gear Box Mill Pulverizer

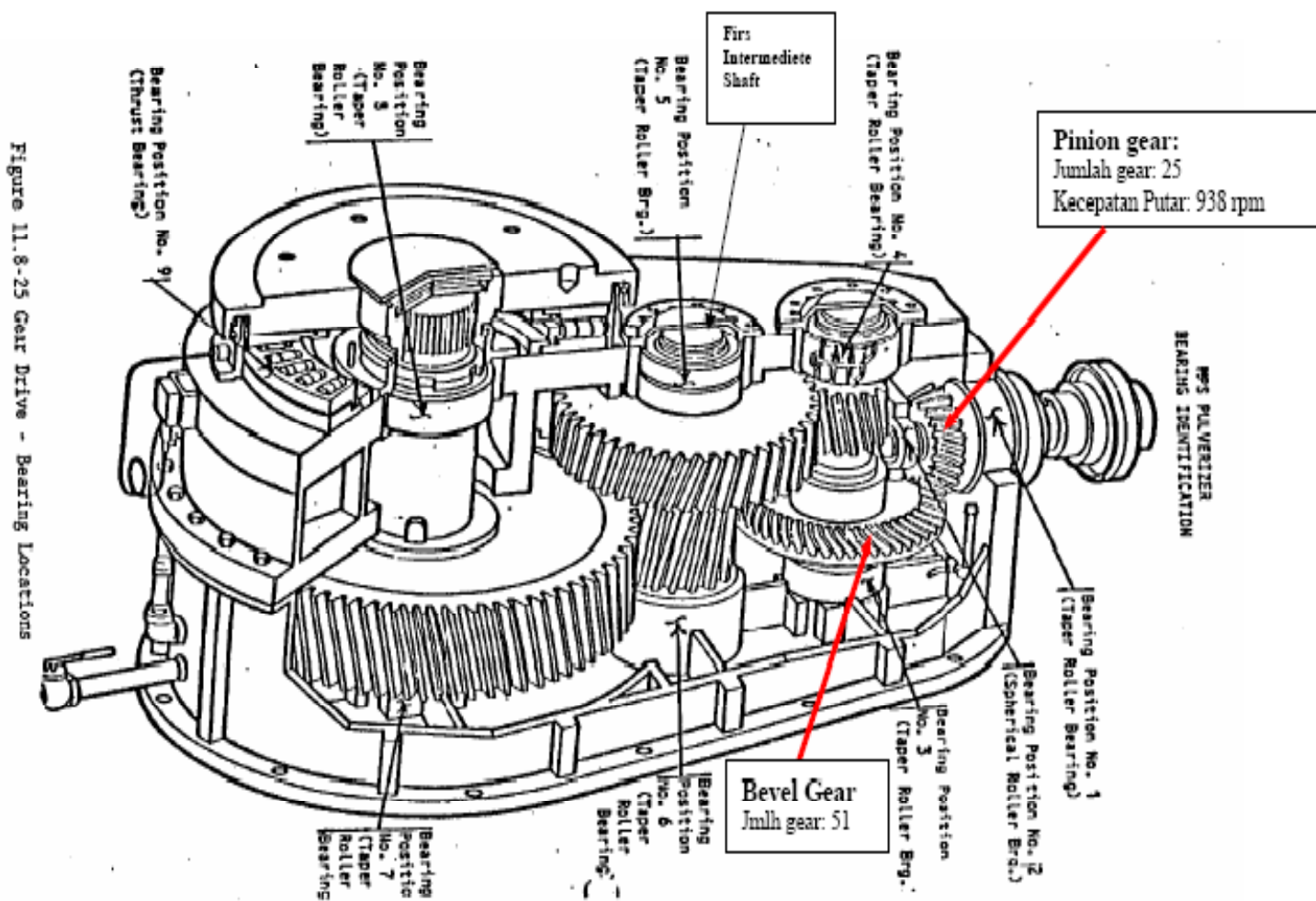


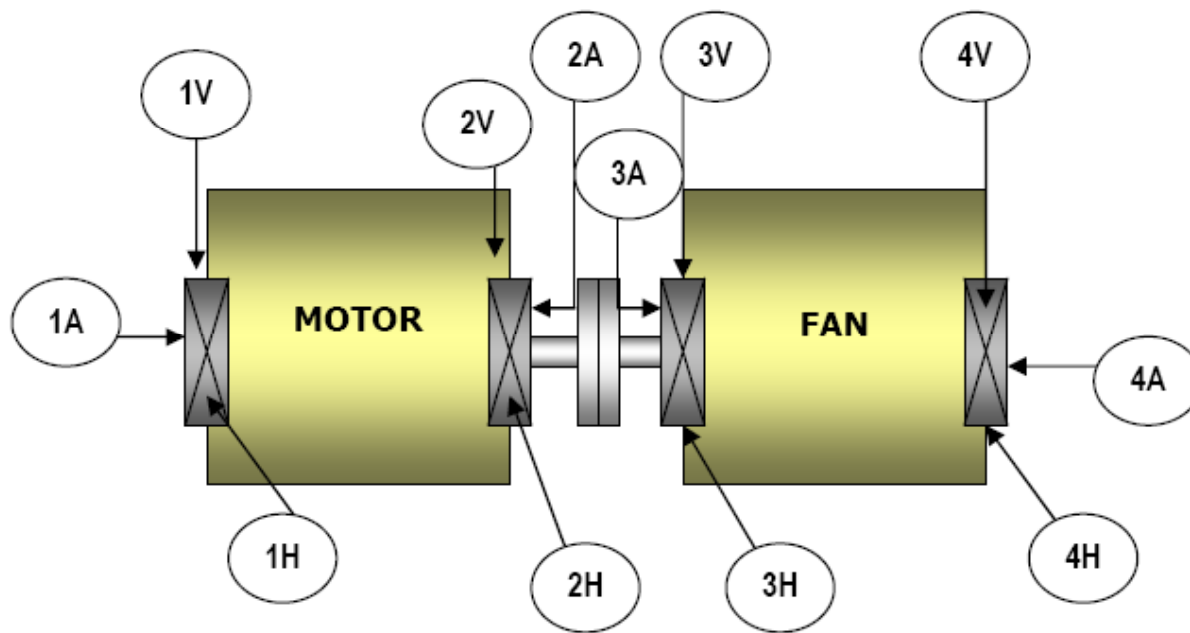
Figure 11.8-25 Gear Drive - Bearing Locations

Gambar Gear Box Pulverizer MPS 89

CONTOH KASUS SHAFT MOTOR PATAH AKIBAT MINYAK PELUMAS



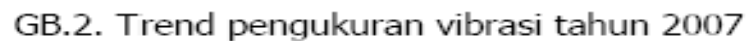


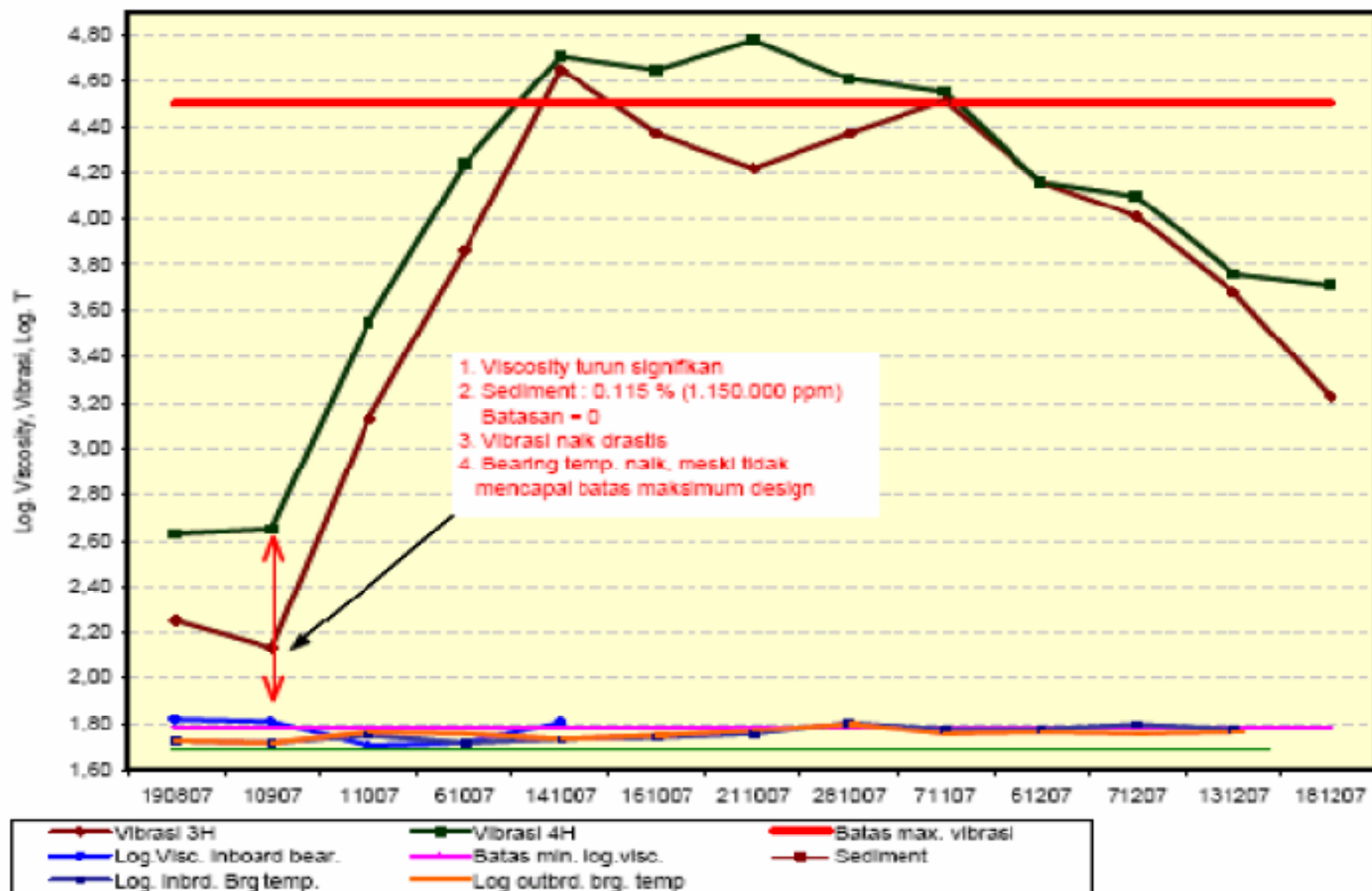


GB.1. Skema titik pengukuran vibrasi

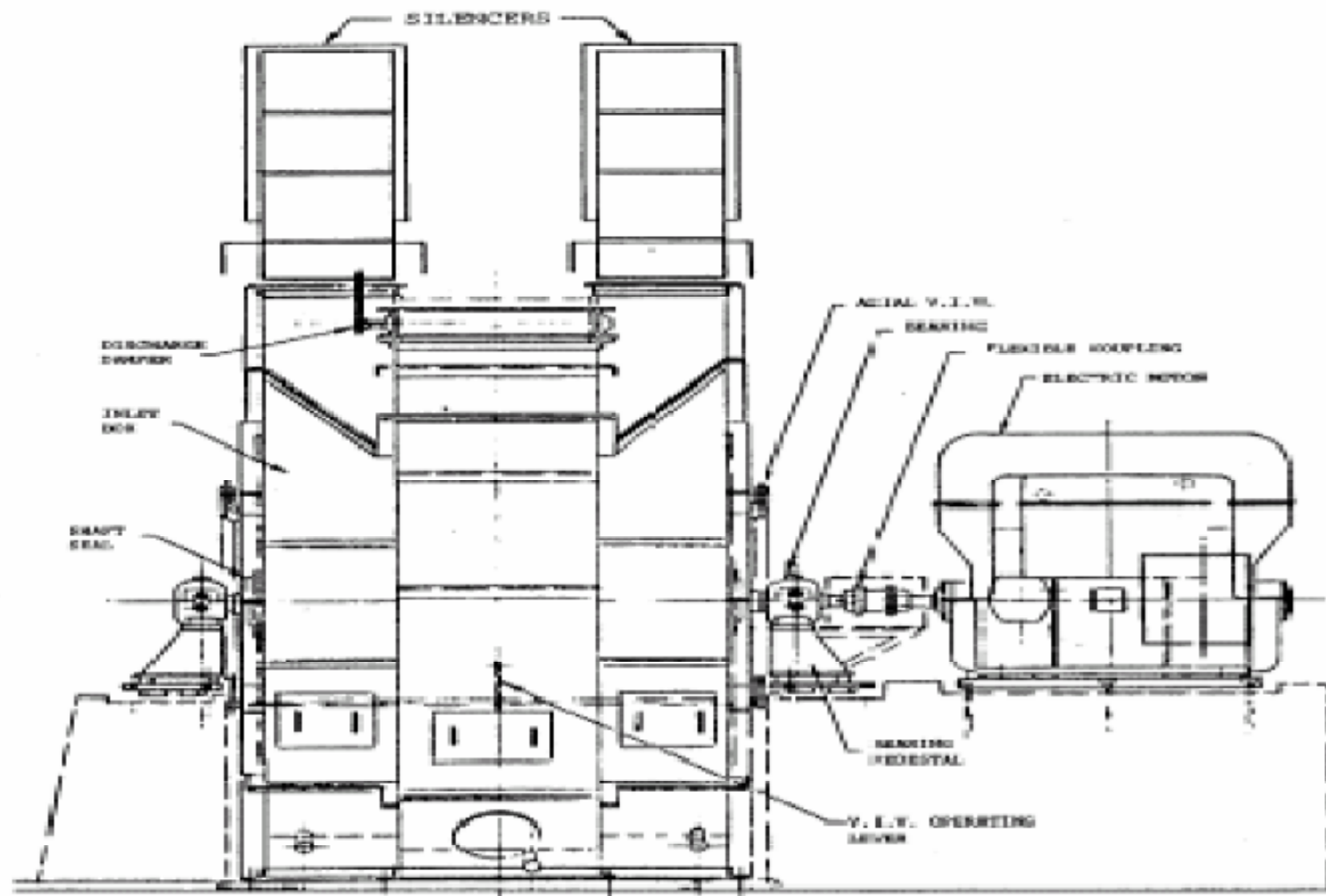


Gb.6. Fan bearing casing retak





Gb.5. Kurva gabungan vibrasi, log. Viscosity dan log temperature.



REFERENSI

1. Drew Trojer & Jim fitch : “Oil Analysis Basic”, Noria Corporation, USA-1999
2. Engineering Tribology : “Gweedon W . Stachowiak & Andrew W. Batchelor, Butterworth
3. NTS Application Notes : “Lubricant Cleanliness”, www.natrib.com
4. “Oil Analysis technology For PDM”, PT.TVP
5. “Lubricant and Lubrication”, www.sainz.com

Dynamic viscosity

Dynamic viscosity is the lubricant property involved in tribological calculations. It provides a relationship between the shear stress and the rate of shear which may be expressed as:

Shear stress

$$= \text{Coefficient of Dynamic Viscosity} \times \text{Rate of Shear}$$

or

$$\tau = \eta \frac{\partial u}{\partial y} = \eta D,$$

where τ = shear stress,

η = dynamic viscosity,

$$\frac{\partial u}{\partial y} = D = \text{rate of shear.}$$

For the parallel-plate situation illustrated in Fig. 6.1.

$$\frac{\partial u}{\partial y} = \frac{U}{h}$$

and

$$\tau = \eta \frac{U}{h}$$

If τ is expressed in N/m^2 and $\frac{\partial u}{\partial y}$ in s^{-1}

then η is expressed in Ns/m^2 , i.e. viscosity in SI units.

The unit of dynamic viscosity in the metric system is the poise ($\frac{\text{g}}{\text{cm s}}$):

$$1 \frac{\text{Ns}}{\text{m}^2} = 10 \text{ poise.}$$

Kinematic viscosity

Kinematic viscosity is defined as $\nu = \frac{\eta}{\rho}$ where ρ is the density of the liquid.

If ρ is expressed in kg/m^3 , then ν is expressed in m^2/s , i.e. in SI units.

The unit of kinematic viscosity in the metric system is the stoke ($\frac{\text{cm}^2}{\text{s}}$).

$$1 \frac{\text{m}^2}{\text{s}} = 10^4 \text{ stokes.}$$

Table 1 gives the factors for converting from SI to other units.

Table 1 Viscosity conversion factors

Dynamic viscosity (SI unit is $\frac{\text{Ns}}{\text{m}^2}$)	
$\frac{\text{Ns}}{\text{m}^2} \times 10$	= poise ($\frac{\text{g}}{\text{cm.s}}$)
poise $\times 0.1$	= $\frac{\text{Ns}}{\text{m}^2}$

Kinetic viscosity (SI unit is $\frac{\text{m}^2}{\text{s}}$)	
$\frac{\text{m}^2}{\text{s}} \times 10^4$	= stokes ($\frac{\text{cm}^2}{\text{s}}$)
stokes $\times 0.1^{-4}$	= $\frac{\text{m}^2}{\text{s}}$